Online CV Management & Search System

Course: Web Programming

Group Size: 3 or 4 students

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1. Assignment Overview

In this assignment, students will design and implement a complete **Online CV Management and Search System**.

The system focuses on:

- Allowing **job seekers** to create and manage a **comprehensive online CV**
- Allowing **employers** to **search, filter, and view CVs**
- Emphasizing **structured data modeling**, **database normalization**, **dynamic web forms**, and **search-ready system design**

This assignment **does NOT include**:

- Job postings
- Job applications
- Matching or recommendation algorithms

These features will be addressed in a **subsequent assignment**.

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2. Learning Outcomes

By completing this assignment, students will be able to:

- Design normalized relational databases
- Implement dynamic, data-driven web forms
- Model one-to-many and many-to-many relationships
- Apply MVC (or equivalent) architecture
- Enforce data consistency using reference tables
- Build role-based web applications
- Design systems optimized for searching and filtering

3. System Roles

3.1 Job Seeker

- Creates and manages one online CV

3.2 Employer

- Searches and views CVs (read-only)

3.3 Administrator

- Manages users and system reference data

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4. Functional Requirements

4.1 Job Seeker Features

4.1.1 Authentication

- Secure registration and login
- Role-based access control

4.1.2 Online CV Creation & Management (Core Requirement)

Each job seeker may create **one (1) complete online CV**.
All CV data must be **stored in the database** using **structured and normalized fields**.

A. Personal Information

Field	Input Type
Full Name	Text
Date of Birth	Date
Gender	Select
Email	Text
Phone Number	Text

B. Structured Contact Address (MANDATORY)

The contact address **must NOT be stored as a single text field**.

To support searching, filtering, and future system expansion, the address must be **split into multiple structured fields**, each stored in a separate database column.

Required Address Fields

Field	Input Type
Country	Select (predefined list)
City / Province	Select (predefined list)
District	Select (optional)
Street Address	Text
Postal Code	Text (optional)

Explicit Constraint

❌ Storing the address as:

- One string
- A textarea
- A JSON field

is **NOT allowed**.

Each address component must be **queryable independently**.

C. CV Category (MANDATORY)

- Each CV must belong to **exactly one predefined CV category**
- Categories represent professional domains

Examples:

- Software Development
- Data Science
- Finance & Accounting
- Marketing
- Education
- Design & Creative

❌ Free-text category input is **not allowed**.

D. Education (Multiple Degrees – Dynamic Form)

- Job seekers must be able to add **multiple education records**
- An **“Add Degree”** button is required

Field	Input Type
Institution	Select / Autocomplete
Degree Level	Select
Major / Field of Study	Select
Start Year	Select
End Year	Select
Description	Optional text

E. Work History (Multiple Records – Dynamic Form)

- Job seekers must be able to add **multiple work history records**
- An **“Add Work History”** button is required

Field	Input Type
Company Name	Text
Job Title	Select
Employment Type	Select
Industry	Select
Start Year	Select
End Year	Select or “Present”
Job Description	Textarea

F. Certificates (Multiple Records – Dynamic Form)

- Job seekers must be able to add **multiple certificates**
- An **“Add Certificate”** button is required

Field	Input Type
Certificate Name	Select
Issuing Organization	Select
Year Issued	Select
Description	Optional text

G. Skills (Strongest Skills Only)

- Each job seeker may define **a maximum of 5 strongest skills**
- The system must enforce this limit at:
  - UI level
  - Server-side validation

Field	Input Type
Skill	Select
Proficiency Level	Select

❌ Free-text skills are **not allowed**.

4.2 Employer Features (Search & View Only)

Employers must be able to:

4.2.1 Search CVs Using Multiple Criteria

- Keyword search (name, summary, descriptions)
- CV category
- Location (country, city)
- Skills (one or more)
- Minimum skill proficiency
- Degree level

4.2.2 Combined Filters

- Multiple filters must be combinable using **AND logic**

4.2.3 Sorting Options

- Most recently updated CV
- Alphabetical order
- Approximate work experience length

4.2.4 CV Viewing

- View CVs in a structured format
- Switch between different CV templates
- Read-only access only

4.3 CV Presentation Templates (MANDATORY)

- The system must provide **at least three (3) distinct CV view templates**
- Templates must:
  - Differ clearly in layout and visual style
  - Use the same underlying CV data
- CV data must be stored **once only**

Examples:

- Modern
- Classic
- Minimal

4.4 Administrator Features

- Manage users (job seekers and employers)
- Manage reference (lookup) tables:
  - Skills
  - CV categories
  - Degrees
  - Majors
  - Industries
  - Locations
  - Employment types
  - Certificates
- Remove invalid or inappropriate data

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5. Database & Data Requirements

- Use a **relational database**
- All CV data must be **fully normalized**
- Reference tables must be used for all selectable fields
- CV-related tables must store **foreign keys only**

❌ Storing structured CV data as:

- Plain text blobs
- JSON fields
- Comma-separated values

is **NOT allowed**.

6. Technical Requirements

- MVC architecture (or equivalent)
- Dynamic forms implemented using JavaScript
- Server-side validation for all inputs
- Role-based access control
- Responsive UI design
- Version control using Git

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7. Deliverables

Item	Description
Source Code	Backend + frontend
Database Design	ER diagram + SQL schema
Technical Report	Architecture & design decisions
Demo Video	~5 minutes
Git Repository	Commit history required

8. Assessment Criteria

Category	Weight
Online CV completeness & correctness	30%
Database design & normalization	25%
Employer search & filtering	15%
Code structure & architecture	15%
UI & CV templates	10%
Documentation & demo	5%
Total	**100%**

9. Constraints & Notes

- Only **one CV per job seeker**
- Maximum **5 strongest skills**
- No job postings
- No job applications
- No CV file uploads
- No AI-based features required

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10. Academic Integrity

All submitted work must be original. External libraries or frameworks may be used but must be properly documented.